Too early to dismiss Yersinia enterocolitica infection in the aetiology of Graves' disease: evidence from a twin case-control study.

BACKGROUND: Yersinia enterocolitica (YE) infection has long been implicated in the pathogenesis of Graves' disease (GD). The association between YE and GD could, however, also be due to common genetic or environmental factors affecting the development of both YE infection and GD. This potential confounding can be minimized by investigation of twin pairs discordant for GD. AIM: To examine whether YE infection is associated with GD. DESIGN: We first conducted a classical case-control study of individuals with (61) and without (122) GD, and then a case-control study of twin pairs (36) discordant for GD. METHODS: Immunoglobulin (Ig)A and IgG antibodies to virulence-associated Yersinia outer membrane proteins (YOPs) were measured. MAIN OUTCOME MEASURES: The prevalence of YOP IgA and IgG antibodies. RESULTS: Subjects with GD had a higher prevalence of YOP IgA (49%vs. 34%, P = 0.054) and YPO IgG (51%vs. 35%, P = 0.043) than the external controls. The frequency of chronic YE infection, reflected by the presence of both IgA and IgG YOP antibodies, was also higher among cases than controls (49%vs. 33%, P = 0.042). Similar results were found in twin pairs discordant for GD. In the case-control analysis, individuals with GD had an increased odds ratio (OR) of YE infection: IgA 1.84 (95% CI 0.99-3.45) and IgG 1.90 (95% CI 1.02-3.55). In the co-twin analysis, the twin with GD also had an increased OR of YE infection: IgA 5.5 (95% CI 1.21-24.81) and IgG 5.0 (95% CI 1.10-22.81). CONCLUSION: The finding of an association between GD and YE in the case-control study and within twin pairs discordant for GD supports the notion that YE infection plays an aetiological role in the occurrence of GD, or vice versa. Future studies should examine the temporal relationship of this association in more depth.